PS50008A — EDS Assignment

Practice Paper — Spring 2026

Research Methods and Experimental Design

2026-02-23

Total marks: 100 (Section A: 50 marks | Section B: 50 marks) | This is a take-home assignment, not a timed exam.

Read the study description below carefully. Answer **all** questions using the R output provided on the following pages. No coding or hand calculations are required.

The Study. A researcher was interested in the effect of background noise on concentration performance, inspired by Mehta, Zhu and Cheema's (2012) research on ambient noise and cognitive functioning.

The researcher recruited 24 university students and randomly assigned them to one of two conditions: a **silent room** condition or a **background café noise** (70 dB) condition, with 12 participants in each group. Each participant was given a standardised proofreading task consisting of a 500-word passage containing 30 deliberate errors (spelling, grammar, and punctuation mistakes). Participants were instructed to read through the passage carefully and circle as many errors as they could find within a 10-minute time limit. The researcher recorded the number of errors correctly detected by each participant (out of 30). All participants were tested individually in the same laboratory room, with the café noise played through calibrated speakers in the noise condition. Participants were thanked and debriefed.

Data: Number of errors detected (out of 30)

Participant	Silent
1	25
2	19
3	24
4	21
5	27
6	20
7	23
8	18
9	26
10	22
11	20
12	25

Participant	Café Noise
13	21
14	15
15	19
16	22
17	14
18	20
19	17
20	16
21	23
22	18
23	3
24	22

Section A: Statistical Mechanics (50 marks)

Question 1 (5 marks)

State the researcher's experimental hypothesis.

Question 2 (12 marks)

State:

a) the type of experimental design (3 marks)
b) the independent variable (3 marks)
c) the dependent variable (3 marks)
d) the type of data (3 marks)

Question 3 (8 marks)

The researcher used R to produce the following descriptive statistics:

```
> data %>%
+   group_by(condition) %>%
+   summarise(Mean = mean(errors_detected),
+             SD = sd(errors_detected),
+             n = n())

# A tibble: 2 x 4
  condition   Mean    SD     n
  <chr>      <dbl> <dbl> <int>
1 Café Noise  17.5  5.42    12
2 Silent      22.5  2.94    12
```

a) Report the means and standard deviations for each condition in APA format. (4 marks)

b) What do these descriptive statistics suggest about the pattern of results? (4 marks)

Question 4 (10 marks)

The researcher then ran the following inferential test in R:

```
> t.test(errors_detected ~ condition, data = data)

        Welch Two Sample t-test

data:  errors_detected by condition
t = -2.81, df = 16.95, p-value = 0.012
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -8.76 -1.24
sample estimates:
mean in group Café Noise    mean in group Silent
                   17.50                   22.50
```

a) Why is an independent samples t-test (rather than a paired t-test) the appropriate test here? (2 marks)

b) State the t-statistic, degrees of freedom, and p-value. (3 marks)

c) Is the result statistically significant at the $\alpha = .05$ level? Explain your answer. (2 marks)

d) Interpret the 95% confidence interval. What does it tell us? (3 marks)

Question 5 (5 marks)

Using the R output from Questions 3 and 4, write a full APA-style results paragraph reporting both the descriptive and inferential statistics, and stating whether the hypothesis was supported.

Question 6: Data Quality (10 marks)

The researcher also produced the following boxplot to visualise the data:

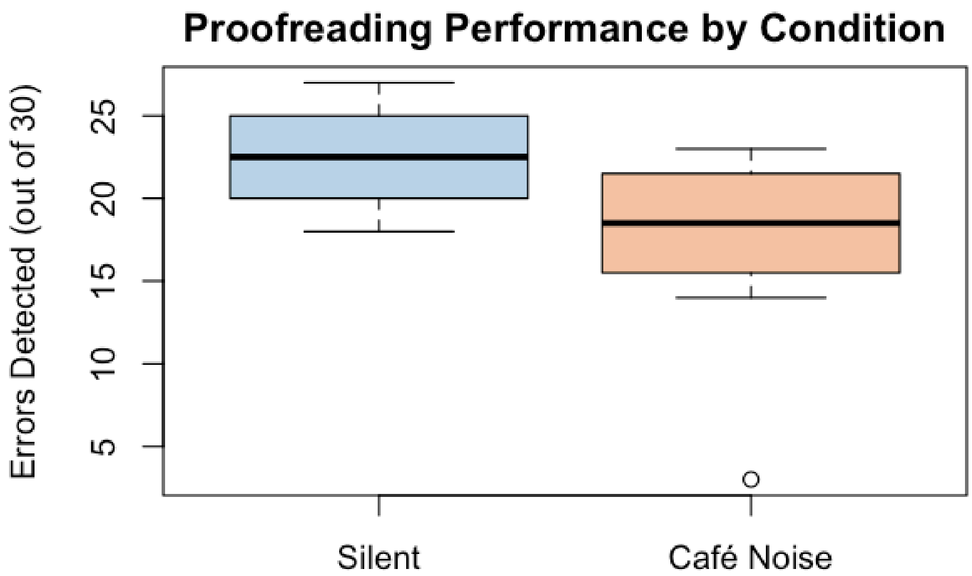

Figure 1: Boxplot of proofreading performance by condition

a) Looking at the boxplot and the data table, identify the unusual data point. Which participant is it, and what did they score? (2 marks)

b) Suggest a plausible reason why this participant's score is so different from the others in their group. (2 marks)

c) Look at the standard deviations reported in Question 3. How does this unusual data point help explain why the SD for the Café Noise group (5.42) is so much larger than the SD for the Silent group (2.94)? (3 marks)

d) What should the researcher do about this data point before drawing conclusions? Justify your answer. (3 marks)

Section B: Methods and Theory (50 marks)

Question 7 (10 marks)

Suggest **two** methodological improvements the researcher could make if the experiment were to be repeated. For each improvement, explain why it would strengthen the study. (5 marks each)

Question 8 (20 marks)

If further research were conducted on the effect of background noise on concentration:

a) Do you think there would be any differences in performance if the results of **introverts** and **extroverts** were compared? Why? (10 marks)

b) Do you think there would be any differences in performance if the results of **university students** and **office workers** were compared? Why? (10 marks)

Question 9 (20 marks)

Outline a more **ecologically valid** way in which the effect of background noise on concentration could be investigated. Your answer should include details of the hypothesis, independent variable (with levels), dependent variable, and procedure.

End of Practice Paper

Model Answers and Marking Guidance

Blue boxes = model answers (what students should write).
Orange boxes = marker notes (guidance for awarding marks).
Grade boundaries: 50+ = Pass | 60+ = Merit | 70+ = Distinction

Section A: Statistical Mechanics (50 marks)

Question 1 (5 marks)

That background noise (café noise at 70 dB) will affect the number of errors detected on a proofreading task compared to working in silence. *OR* That participants in the silent condition will detect more proofreading errors than participants in the café noise condition.

5 marks for a clear hypothesis referencing both IV and DV. **3 marks** if vague (e.g., "noise affects concentration"). **0 marks** for a null hypothesis.

Question 2 (12 marks)

a) Between-subjects / independent groups design.
b) The noise condition — silent room vs background café noise (70 dB).
c) The number of errors correctly detected on the proofreading task (out of 30).
d) Ratio data (discrete).

3 marks each. Accept "independent measures" or "unrelated design" for (a). Must identify both levels for full marks on (b). Accept ratio or interval for (d). Award 0 for "repeated measures" on (a) — different participants in each group.

Question 3 (8 marks)

a) Participants in the silent condition detected more errors (M = 22.50, SD = 2.94) than participants in the café noise condition (M = 17.50, SD = 5.42).

b) The descriptives suggest that participants performed better when working in silence — detecting on average 5 more errors. The much larger SD in the Café Noise group (5.42 vs 2.94) indicates greater variability, suggesting some participants were more affected by noise than others.

4 marks each. For (a): correct values in APA format with italicised *M* and *SD*. Deduct 1 per wrong value. For (b): must link pattern to the research question, not just restate numbers. Award bonus consideration if student notes the SD difference.

Question 4 (10 marks)

a) An independent samples t-test is appropriate because different participants were in each condition (between-subjects design). The scores are not paired.

b) $t = -2.81$, $df = 16.95$, $p = .012$

c) Yes, the result is statistically significant because $p = .012 < .05$. We reject the null hypothesis.

d) The 95% CI [−8.76, −1.24] tells us the true difference between groups is estimated to lie between −8.76 and −1.24 errors. It does not contain zero, confirming a real difference. The entirely negative interval indicates the café noise group detected fewer errors.

(a) 2 marks — must mention different participants / between-subjects. **(b) 3 marks** — 1 each for *t*, *df*, *p*. **(c) 2 marks** — must state $p < .05$ and the conclusion. **(d) 3 marks** — must note CI excludes zero and interpret the direction.

Question 5 (5 marks)

An independent samples *t*-test revealed that participants in the silent condition detected significantly more proofreading errors ($M = 22.50$, $SD = 2.94$) than participants in the café noise condition ($M = 17.50$, $SD = 5.42$), $t(16.95) = -2.81$, $p = .012$, 95% CI [−8.76, −1.24]. This supports the hypothesis that background noise impairs concentration performance.

5 marks for: test named, both descriptives in APA, test statistic with *df* and *p*, conclusion stated. **3 marks** if most elements present but poorly formatted. **1 mark** for conclusion only.

Question 6: Data Quality (10 marks)

a) Participant 23 in the Café Noise condition scored only 3 out of 30. This is the outlier (open circle) below the lower whisker of the boxplot. All other noise-group participants scored between 14 and 23.

b) The participant may not have been engaged with the task, may have misunderstood the instructions (e.g., only circled one type of error), or may have had difficulty reading the passage.

c) The score of 3 is very far from the group mean (17.50) — a deviation of 14.5 points. This single extreme score greatly inflates the SD. Without Participant 23, the noise group SD drops to 3.06 — much closer to the silent group's 2.94. One extreme score can dramatically affect measures of spread.

d) The researcher should: (1) investigate *why* the score is so low — check for procedural issues; (2) run the analysis both with and without the participant and report both for transparency. They should not simply delete it without justification. The result is significant either way ($p = .012$ with, $p = .008$ without), so the conclusion holds.

(a) 2 marks — identify P23 and score of 3. **(b) 2 marks** — any plausible, explained reason. **(c) 3 marks** — must explain the outlier inflates SD because it is far from the mean; bonus for noting the SD drops to 3.06 without it. **(d) 3 marks** — must discuss investigation + reporting both analyses for transparency. Award 1 mark for "just remove it" without justification.

Section B: Methods and Theory (50 marks)

Question 7 (10 marks)

Any two well-justified improvements. Examples:

1. **Increase sample size** — 12 per group is small; a larger sample increases statistical power and generalisability.
2. **Control for hearing ability** — participants were not screened for hearing difficulties, which could confound results.
3. **Standardise noise exposure** — ensure all noise-condition participants hear the same clip; natural café noise varies.
4. **Use multiple proofreading tasks** — a single passage may not be representative; several texts increases reliability.
5. **Control for time of day** — cognitive performance varies across the day; testing at similar times reduces this confound.
6. **Screen for engagement** — given the outlier (P23), include a minimum performance criterion.

5 marks each if clearly stated and well justified. **3 marks** if identified but poorly justified. **1 mark** if vague with no explanation. Must be *two different* improvements.

Question 8 (20 marks)

a) Introverts vs Extroverts: According to Eysenck's arousal theory, introverts have naturally higher cortical arousal. They are more sensitive to external stimulation — café noise may push them beyond their optimal arousal level, causing greater impairment. Extroverts, with lower baseline arousal, may be less affected or may even benefit from moderate noise. Prediction: introverts would show a larger drop in performance.

b) University Students vs Office Workers: Students often study in cafés, libraries with background chatter, and shared accommodation — they may be more habituated to working with noise. Office workers may be used to quieter environments. The key argument is about habituation and adaptation — the more accustomed group may be less affected. Could also argue office workers have developed stronger concentration strategies through professional experience.

10 marks each. Award full marks for well-reasoned answers with theoretical or empirical support (e.g., Eysenck for 8a). **5–7 marks** for reasonable argument without theoretical backing. **1–3 marks** for vague assertions. Either direction is acceptable if well argued.

Question 9 (20 marks)

Hypothesis: Students who study in a café will recall fewer key concepts from a textbook chapter than students who study in a quiet library.

IV and levels: Study environment — university library (quiet) vs campus café (natural background noise). More ecologically valid because participants experience real noise in real study settings.

DV: Score on a short comprehension test based on the material studied.

Procedure: Participants study a textbook chapter for 20 minutes in their assigned location, then complete a comprehension test in a neutral room. Control for time of day and difficulty of material.

Why more ecologically valid: Uses real environments where students actually study, with natural noise rather than recorded/calibrated noise, and a genuine study task rather than artificial lab conditions.

15–20 marks for detailed proposal with hypothesis, IV, DV, procedure, and ecological validity justification. **10–14 marks** for reasonable proposal with most elements. **5–9 marks** for basic suggestion. **1–4 marks** for one-line answer. Increase marks for those who outline design aspects in detail.

End of Practice Paper with Model Answers